

REPORT OF WEEK 5,6

Project: Digital Detective

WEEK 5: ENEMY AND COMBAT / INTERACTION SYSTEM

During Week 5, the main focus was to improve the gameplay of the Digital Detective project by adding character movement and interaction features. The basic game environment was already prepared, so this week was mainly used to make the detective character more interactive and to connect the player with different objects in the game.

1. Detective Character Development

- The detective character was improved so that the player can control the character inside the game area.
- A simple movement system was added to make the detective move in different directions.
- The character is represented by a blue box in the current version of the game.
- The movement was designed to be simple because the main purpose of the project is investigation rather than fast action gameplay.

2. Player Movement System

- A keyboard-based movement system was added for the detective.
- The player can use the **W, A, S and D keys** to move the character around the investigation area.
- The movement boundaries were added so that the detective stays inside the game environment.
- This prevents the character from going outside the playable screen.
- The movement system was tested to make the controls easier for the player.

3. Investigation System

- An investigation system was added to allow the detective to interact with important objects in the case.
- Different investigation points were placed inside the game area.
- When the detective reaches a clue or investigation point, the player can press the **E key** to investigate it.
- This makes the game more interactive because the player needs to explore the environment and find important objects.
- The system connects the player's movement with the main investigation activities.

4. Evidence Collection

- An evidence collection feature was developed as part of the investigation system.
- Different clues were placed around the game area for the detective to discover.
- The player can approach a clue and use the interaction key to collect it.
- After collecting a clue, it is added to the evidence list.
- The player's score is also updated after collecting evidence.
- A message is displayed to inform the player when new evidence has been found.

5. NPC Interaction

- An NPC interaction system was added to make the investigation more realistic.
- The detective can approach the NPC inside the game environment.
- By pressing the interaction key, the player can start a conversation with the NPC.
- The dialogue provides information related to the missing diamond case.
- The information from the NPC helps the player understand the situation and identify possible suspects.

6. Dialogue System

- A simple dialogue system was added for communication between the detective and the suspect.
- The dialogue is displayed through message boxes when the player interacts with the NPC.
- The conversation contains basic questions and answers related to the case.
- Important information from the conversation can be used later during the investigation.
- This feature connects the investigation system with the main story of the game.

7. Score System

- A simple scoring system was included to make the investigation more interesting.
- The player receives points after collecting evidence.
- Points can also be added after completing important investigation activities.
- The current score is displayed on the game screen.
- This allows the player to easily track their progress during the case.

8. Testing and Improvement

- The movement and interaction features were tested after implementation.
- The detective's movement was checked using the W, A, S and D keys.
- Evidence collection and NPC interaction were also tested.
- Some movement and keyboard control problems were identified during testing.
- The controls were adjusted to make the gameplay easier and more stable.

WEEK 6: INVESTIGATION, PUZZLE AND CASE COMPLETION

During Week 6, the main focus was to develop the investigation part of the Digital Detective game further. The evidence collection and NPC interaction from Week 5 were connected with a puzzle and case completion system. This made the game feel more like an actual investigation game instead of only a basic movement demo.

1. Advanced Investigation System

- The investigation system was improved so that different activities could be connected to the same case.
- The detective needs to explore the area and interact with different investigation points.
- The player can find evidence, talk to the suspect and interact with the puzzle area.
- These activities are connected to the main story of the missing diamond case.
- This creates a simple investigation process for the player to follow.

2. Clue and Evidence System

- Several clues were added to the investigation environment.
- The clues are displayed as question-mark objects so that the player can identify possible investigation points.
- The detective can move towards a clue and press **E** to investigate it.
- When a clue is collected, it is removed from the active investigation area.
- The evidence counter and score are updated after collecting the clue.

3. Evidence Information

- A message is displayed when evidence is successfully collected.
- The message informs the player about the type of evidence that has been found.
- The collected evidence is counted on the game screen.
- This allows the player to track the progress of the investigation.
- The evidence system also gives the player a reason to explore different areas of the game.

4. Puzzle Development

- An interactive puzzle was added to make the investigation more challenging.
- The puzzle is connected to the information received from the suspect.
- It is also related to the evidence found during the investigation.
- The player needs to read the given information and enter an answer.
- This adds a small problem-solving part to the game.
- The puzzle makes the gameplay more engaging instead of only collecting objects.

5. Answer Checking System

- An answer-checking system was added to the puzzle.
- The player can enter an answer using the input box.
- The game checks whether the entered answer is correct or incorrect.
- If the answer is correct, the puzzle is marked as completed.
- Additional points are added to the player's score after solving the puzzle.
- If the answer is wrong, a message is displayed and the player can try again.

6. NPC and Story Information

- The NPC dialogue was connected more closely with the investigation.
- The suspect provides information about the night of the incident.
- The suspect also mentions an important detail about a red thread.
- This information becomes useful when solving the puzzle.
- The dialogue helps the player understand the connection between the suspect and the crime scene.
- This makes the story more connected to the gameplay.

7. Case Progress System

- A case progress system was developed to keep track of the player's investigation.
- The game checks whether the required clues have been collected.
- It also checks whether the NPC has been interviewed.
- The puzzle must also be solved before the case can be completed.
- The case does not finish randomly before the required activities are completed.
- This gives the player a clear objective during the investigation.

8. Case Completion Screen

- A case completion screen was added after the investigation is successfully completed.
- A "**CASE SOLVED!**" message is displayed to show that the case has been completed.
- The final score is displayed on the completion screen.
- A message is also shown to inform the player that the interactive story is complete.
- This provides a clear ending to the current investigation case.

9. Restart Feature

- A restart feature was added so that the player can play the case again.
- The player can press **ENTER** after completing the case.
- A new game starts from the beginning.
- The previous score is reset.
- Collected evidence is also cleared.
- The puzzle and NPC progress are reset for the new game.

10. Final Testing

- The complete Week 5 and Week 6 features were tested together.
- The detective movement system was checked to make sure the character could move around the game area.
- Evidence collection was tested with different clues.
- NPC conversation and dialogue were tested.
- The puzzle and answer-checking system were also tested.
- The score and case completion system were checked after completing the required tasks.
- Movement and keyboard control issues found during testing were adjusted to provide smoother gameplay.